

SYDNEY TECHNICAL HIGH SCHOOL



Mathematics Extension 2

**HSC ASSESSMENT TASK 1
MARCH 2008**

General Instructions

- Working time allowed – 70 minutes
- Write using black or blue pen
- Approved calculators may be used
- All necessary working should be shown
- Start each question on a new page
- Attempt all questions
- All questions are of equal value

NAME : _____

QUESTION 1	QUESTION 2	QUESTION 3	TOTAL

Question 1 (16 marks)

a) If $z = 1 - \sqrt{3}i$ 5

i) find $|z|$

ii) find $\arg(z)$

iii) find z^5 in the form $a + ib$

iv) find a possible value of n ($n > 1$) such that $\arg(z) = \arg(z^n)$

b) Solve the equation : $z^2 + 4z - 1 + 12i = 0$. 4

c) Find the equation of the ellipse with eccentricity $\frac{4}{5}$ 2

and foci at $(-8, 0)$ and $(8, 0)$.

d) Find the gradient of the tangent to $4x + xy^2 = y^3$ at the point $(1, 2)$. 2

e) The equation of the normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 3

at the point $P(x_1, y_1)$ is given by $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$.

This normal meets the minor axis of the ellipse at G .

The line parallel to the major axis of the ellipse which passes

through the point P meets the minor axis of the ellipse at N .

Show that $\frac{OG}{ON} = \frac{-a^2e^2}{b^2}$

Question 2 (16 marks)

- a) Sketch on an Argand diagram the region specified by 2

$$|z - 2| < |z + 2i|$$

- b) i) Sketch the locus of the point z such that $|z - (3 + 2i)| = 2$ 2

- ii) Determine the values of k for which the simultaneous equations 2

$$|z - 2i| = k \quad \text{and} \quad |z - (3 + 2i)| = 2 \quad \text{have exactly two solutions.}$$

- c) On an Argand diagram the quadrilateral OABC is a square, 2

where O is the origin.

If A represents the complex number $5 + 2i$ find the complex number

represented by the points B and C given that they both have positive arguments.

- d) i) Solve $z^3 = 1$ over the complex field. 2

- ii) Given that ω is the complex roots of $z^3 = 1$ with smallest positive argument :

α) Show that $1 + \omega + \omega^2 = 0$ 1

β) Evaluate $(1 - \omega)(1 - \omega^2)(1 - \omega^4)(1 - \omega^8)$ 3

- e) Use De Moivre's Theorem to show that 2

$$(\cot \theta + i)^n + (\cot \theta - i)^n = \frac{2 \cos n\theta}{\sin^n \theta}$$

Question 3 (16 marks)

- a) Given that z is a complex number, show that the locus defined by 3

$$z\bar{z} + 10(z + \bar{z}) = 21$$

is a circle and state its centre and radius.

- b) Sketch the locus of the point z such that $\arg(z + 2i) = \frac{3\pi}{4}$ 2

- c) Sketch the locus of z given that $\frac{z}{z+4}$ is purely imaginary. 2

- d) i) Show that the equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 3

at the point $P(a \cos \theta, b \sin \theta)$ is given by $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$

- ii) If this tangent at P meets the x axis at A 4

and S is a focus of the ellipse

show that $\frac{PS}{AS}$ is independent of the values of a and b .

- e) Given that $|z-2|=2$ and $0 < \arg(z) < \frac{\pi}{2}$ 2

find the value of k if $\arg(z-2) = k \times \arg(z^2 - 2z)$

The End

Question

$$r = \frac{11}{3}$$

$$\text{iii) } z^5 = \left[2 \operatorname{cis} \left(\frac{\pi}{3} \right) \right]^5$$

$$= 32 \operatorname{cis} \left(\frac{5\pi}{3} \right)$$

$$= 16 + 16\sqrt{3}i$$

$$\text{iv) } n=7$$

$$\text{b) } z = \frac{-4 \pm \sqrt{16 - 4(1)(-1+12i)}}{2}$$

$$= -2 \pm \sqrt{5-12i}$$

$$\sqrt{5-12i} = a+ib$$

$$5-12i = a^2 - b^2 + i(2ab)$$

$$\therefore a^2 - b^2 = 5$$

$$ab = -6$$

$$\therefore a=3, b=-2$$

$$\therefore z = -2 \pm (3-2i) \\ = -5+2i, 1-2i$$

$$\begin{aligned} \text{c) } ae &= 8 & b^2 &= a^2(1-e^2) \\ a \cdot \frac{4}{5} &= 8 & &= 10^2 \left(1 - \frac{16}{25} \right) \\ a &= 10 & &= 36 \\ & & b &= 6 \end{aligned}$$

$$\therefore \frac{x^2}{100} + \frac{y^2}{36} = 1$$

$$1 + 2y^2 = 0 \Rightarrow \frac{dy}{dx} = \frac{1}{2y} \cdot \frac{dy}{dx}$$

$$(3y^2 - 2xy) \frac{dy}{dx} = 4 + y^2$$

$$\frac{dy}{dx} = \frac{4+y^2}{3y^2-2xy}$$

$$\text{sub (1,2)}$$

$$m_T = \frac{4+4}{12-4} \\ = 1$$

$$\text{e) } N(0, y_1)$$

when $x=0$ normal becomes

$$-\frac{b^2 y}{y_1} = a^2 - b^2$$

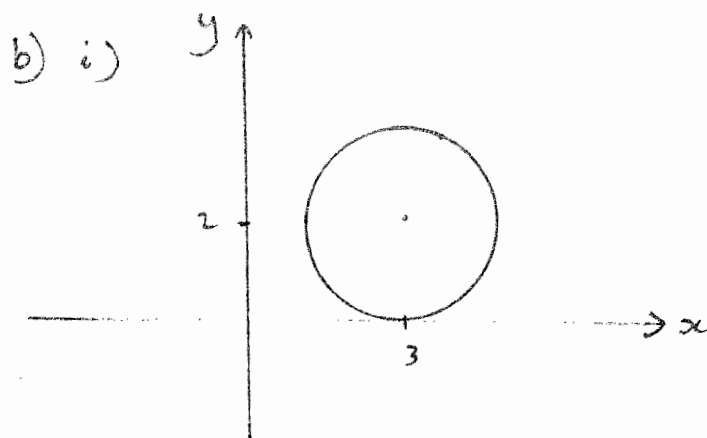
$$y_1 = \frac{y_1}{-b^2} (a^2 - b^2)$$

$$\therefore G\left(0, \frac{y_1}{-b^2} (a^2 - b^2)\right)$$

$$\therefore \frac{OG}{ON} = \frac{\frac{y_1}{-b^2} (a^2 - b^2)}{y_1} \\ = \frac{a^2 - b^2}{-b^2}$$

$$\begin{aligned} &= \frac{-a^2 e^2}{b^2} \quad \text{but } b^2 = a^2(1-e^2) \\ & \quad e^2 = 1 - \frac{b^2}{a^2} \\ & \quad = \frac{a^2 - b^2}{a^2} \\ \therefore a^2 e^2 &= a^2 - b^2 \end{aligned}$$

Question 1



ii) $1 < k < 5$

c) $C \rightarrow i(5+2i)$
 $= -2 + 5i$

$B \rightarrow (-2+5i) + (5+2i)$
 $= 3+7i$

d) i) $z = 1, \text{cis } \frac{2\pi}{3}, \text{cis } \frac{4\pi}{3}$

ii) $1, \text{cis } \frac{2\pi}{3}, \text{cis } (-\frac{2\pi}{3})$

iii) $1, \frac{1}{2}(-1+\sqrt{3}i), \frac{1}{2}(-1-\sqrt{3}i)$

1. $z^3 = 1$

2. $z^3 - 1 = 0$

sub $z = w$

$(w-1)(w^2+w+1) = 0$

but $w-1 \neq 0$ as w is complex root

$\therefore w^2 + w + 1 = 0$

iii) $(1-w)(1-w^2)(1-w^4)(1-w^8)$

$= (1-w)^2 (1-w^2)^2$

$= (1-2w+w^2)(1-2w^2+w^4)$

$= (-3w)(-3w^2)$

$= 9w^3$

$= 9$

e) LHS $= (\cot \theta + i)^n + (\cot \theta - i)^n$

$= \left(\frac{\cos \theta + i \sin \theta}{\sin \theta} \right)^n + \left(\frac{\cos \theta - i \sin \theta}{\sin \theta} \right)^n$

$= \frac{\cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta}{\sin^n \theta}$

$= \frac{2 \cos n\theta}{\sin^n \theta}$

$= \text{RHS}$

Ex 10.4 Q 1

$$x+iy = 10 + i(-10) \Rightarrow x+iy = 10-10i$$

$$x^2 + y^2 + 20x = 11^2$$

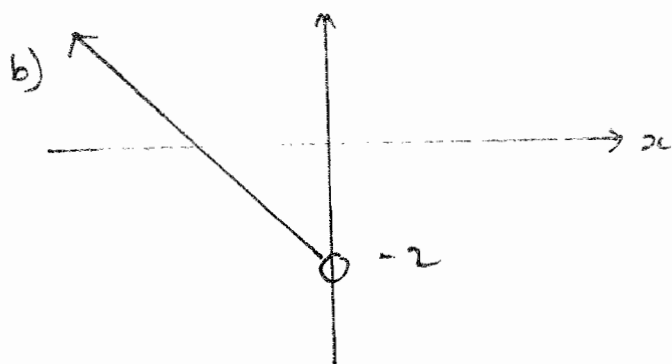
$$x^2 + 20x + 100 + y^2 = 121$$

$$(x+10)^2 + y^2 = 11^2$$

which is a circle

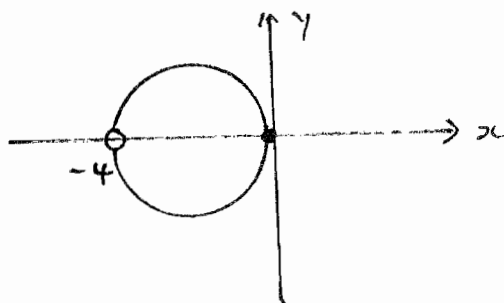
radius 11 units

centre $(-10, 0)$



c) purely imaginary

$$\therefore \arg\left(\frac{z}{z+4}\right) = \pm \frac{\pi}{2}$$



$$\begin{aligned} \text{or } \frac{z}{z+4} &= \frac{x+iy}{x+4+iy} \times \frac{x+4-iy}{x+4-iy} \\ &= \frac{x(x+4)+y^2 + i(y(x+4)-xy)}{(x+4)^2 + y^2} \end{aligned}$$

purely imaginary $\Rightarrow$ real part is zero

$$\frac{dx}{dt} = \frac{b^2}{a^2} \frac{dy}{dt}$$

$$\frac{dy}{dx} = \frac{b^2}{a^2} \frac{1}{x}$$

$$y = \frac{-b^2 a \cos \theta}{a^2 b \sin \theta}$$

$$= \frac{-b \cos \theta}{a \sin \theta}$$

$\therefore$ equation of tangent

$$y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$ay \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$$

$$bx \cos \theta + ay \sin \theta = ab$$

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

$$ii) PS = e PN$$

$$= e \left(\frac{a}{e} - a \cos \theta \right)$$

$$= a - ae \cos \theta$$

$$\text{when } y=0 \quad \frac{x \cos \theta}{a} = 1$$

$$x = \frac{a}{\cos \theta}$$

$$\therefore A \left(\frac{a}{\cos \theta}, 0 \right)$$

$$\therefore AS = \frac{a}{\cos \theta} - ae$$

$$= \frac{a - ae \cos \theta}{\cos \theta}$$

$$\therefore \frac{PS}{AS} = \frac{a - ae \cos \theta}{\frac{a - ae \cos \theta}{\cos \theta}}$$

$$= \cos \theta$$



$$\text{let } \arg(z) = \alpha$$

$$\therefore \arg(z^2) = 2\alpha$$

$$\begin{aligned} \arg(z^2) &= k \times \arg(z^2 - 2z) \\ &= k \times [\arg(z) + \arg(z-2)] \end{aligned}$$

$$\therefore 2\alpha = k [\alpha + 2\alpha]$$

$$k = \frac{2}{3}$$